

ASSIGNMENT NO. 2

PROBLEM STATEMENT:

Write a menu driven program in C to implement the basic arithmetic operations such as Addition, Subtraction, Multiplication, and Division.

OBJECTIVES:

- To understand the concept of menu driven programming in C.
- To apply conditional statements and switch–case control structures.
- To perform basic arithmetic operations using user-defined input.
- To enhance logical thinking and problem-solving skills.

THEORY:

if–else ladder statements syntax and sample program

switch–case or if–else ladder statement syntax and sample program

Basic arithmetic operations include Addition, Subtraction, Multiplication, and Division. Division by zero should be avoided.

CODE:

```
#include <stdio.h>

int main() {
    int choice;
    double num1, num2, result;

    while (1) {
        // Display menu
        printf("\n=== Arithmetic Calculator ===\n");
        printf("1. Addition\n");
        printf("2. Subtraction\n");
        printf("3. Multiplication\n");
        printf("4. Division\n");
        printf("5. Exit\n");
        printf("Enter your choice (1-5): ");
        scanf("%d", &choice);

        if (choice == 5) {
            printf("Goodbye!\n");
            break;
        }

        if (choice < 1 || choice > 5) {
            printf("Invalid choice! Please try again.\n");
            continue;
        }

        // Input numbers
        printf("Enter two numbers: ");
        scanf("%lf %lf", &num1, &num2);

        switch (choice) {
            case 1:
                result = num1 + num2;
                printf("Result: %.2lf\n", result);
                break;
            case 2:
                result = num1 - num2;
                printf("Result: %.2lf\n", result);
                break;
            case 3:
                result = num1 * num2;
                printf("Result: %.2lf\n", result);
                break;
            case 4:
                if (num2 != 0) {
                    result = num1 / num2;
                    printf("Result: %.2lf\n", result);
                } else {
                    printf("Error: Division by zero!\n");
                }
                break;
        }

        return 0;
    }
}
```

OUTPUT:

```
=== Arithmetic Calculator ===
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 1
Enter two numbers: 2 8
Result: 10.00
```

```
=== Arithmetic Calculator ===
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 2
Enter two numbers: 50 44
Result: 6.00
```

```
=== Arithmetic Calculator ===
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 3
Enter two numbers: 33 10
Result: 330.00
```

```
=== Arithmetic Calculator ===
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 4
Enter two numbers: 444 4
Result: 111.00
```

```
=== Arithmetic Calculator ===
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 5
Goodbye!
```

CONCLUSION:

Thus, the menu driven program in C successfully performs basic arithmetic operations using switch–case statements.